Classical Field Theory — Exercise Sheet 21

Vector mass matrices and longitudinal energy

Hints and answers

Use the signature $(+,-,-,-)$ and canonically normalised real scalar kinetic coefficient $1/2$. The questions concern quadratic theories near a fixed vacuum. They make no assertion about a global unitary gauge.

Exercise 21.1 (An orbit determines a vector mass matrix). Let a compact Lie group act orthogonally on $\mathbb{R}^N$, and choose a basis of its Lie algebra, represented by skew real generators T_a, $1 \le a \le r$. Let u_0 be a vacuum of an invariant C^2 potential. Fix $g > 0$, and give the scalar the kinetic term

$$\frac{1}{2}(D_\mu\phi)\cdot(D^\mu\phi), \qquad D_\mu\phi = \partial_\mu\phi + gA^a_\mu T_a\phi.$$

The vector kinetic terms are canonical at the quadratic level.

(a) Derive the vector squared-mass matrix at $\phi = u_0$. Prove that it is nonnegative and identify its kernel and rank. Include generators which act trivially in the representation.

(b) For the three-dimensional rotation representation, take

$$T_a w = e_a \times w, \qquad u_0 = (v/\sqrt{2}, v/\sqrt{2}, 0), \qquad v > 0.$$

Compute the full matrix, its orthonormal eigenvectors and the vector masses. Which gauge-field combination remains massless?

Hint. At a constant vacuum, keep the part of $D_\mu\phi$ linear in A. Evaluate its Gram quadratic form on a real parameter array. For the example use $(a\times u)\cdot(b\times u) = (a\cdot b)|u|^2 - (a\cdot u)(b\cdot u)$.

Answer. The squared-mass matrix is $\mathsf{M}_{ab} = g^2(T_a u_0)\cdot(T_b u_0)$. Its kernel is $\{z : \sum_a z^a T_a u_0 = 0\}$ and its rank is $\dim G - \dim H$ for the vacuum stabiliser H. In the example it is

$$g^2v^2\begin{pmatrix} 1/2 & -1/2 & 0 \\ -1/2 & 1/2 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The combination $(A^1 + A^2)/\sqrt{2}$ is massless. The other two vector masses are gv.

Exercise 21.2 (The charged and neutral electroweak combinations). The following bosonic representation data are given. A complex doublet $\Phi \in \mathbb{C}^2$ has kinetic term $\frac{1}{2}(D_\mu\Phi)^\dagger D^\mu\Phi$, potential $\lambda(\Phi^\dagger\Phi - v^2)^2/8$, and vacuum $\Phi_0 = (0, v)^{\mathsf{T}}$. Here $g, g', v, \lambda > 0$ and

$$D_\mu = \partial_\mu - \mathrm{i}\left(gW_\mu^a\frac{\sigma_a}{2} + g'B_\mu\frac{I_2}{2}\right).$$

All four real vector fields have canonical quadratic Maxwell kinetic terms. The Pauli matrices and the hypercharge generator $Y = I_2/2$ specify the representation; no fermion sector is included.

(a) Find the full real squared-mass matrix in the order (W^1, W^2, W^3, B). Diagonalise its neutral block and define Z and the massless photon A_γ with unit kinetic normalisation. Define $W^\pm = (W^1 \mp \mathrm{i}W^2)/\sqrt{2}$ and find their common mass.

(b) Find the generator annihilating Φ_0 and the coefficient of its photon coupling in D_μ. Find the radial scalar mass and compare the quadratic number of physical modes in the uncoupled limit $g, g' \to 0$ with the broken quadratic description at positive couplings. Explain why W^+ and W^- do not count as two independent complex vectors.

This is the bosonic mass calculation of the electroweak model. For the original construction see Weinberg, *Physical Review Letters* **19** (1967), 1264–1266, equations (4), (7)–(12). The signs and scalar normalisation to use are those explicitly given above.

Hint. Apply the covariant derivative to the two entries of Φ_0 before squaring. The neutral scalar combination is $-gW^3 + g'B$. Diagonalise that rank-one real quadratic form by an orthogonal rotation.

Answer. Writing $g_Z = \sqrt{g^2 + g'^2}$,

$$Z = \frac{gW^3 - g'B}{g_Z}, \qquad A_\gamma = \frac{g'W^3 + gB}{g_Z}, \qquad m_W = \frac{gv}{2}, \quad m_Z = \frac{g_Z v}{2}, \quad m_\gamma = 0.$$

The unbroken generator is $\sigma_3/2 + Y$, with photon coupling $e = gg'/g_Z$. The radial mass is $\sqrt{\lambda}\,v$. Both descriptions contain twelve physical quadratic modes.

Exercise 21.3 (The positive reduced Proca Hamiltonian). Use Lecture 21's variables $B = -\beta\,\mathrm{d}t + \mathbf{b}\cdot\mathrm{d}x$, mass $m > 0$ and spatial momentum $\mathbf{\Pi}$. Take real Schwartz data on $\mathbb{R}^3$. Before elimination the Hamiltonian is

$$H_B = \int_{\mathbb{R}^3} \left[\frac{1}{2}|\mathbf{\Pi}|^2 + \frac{1}{2}|\operatorname{curl}\mathbf{b}|^2 + \frac{1}{2}m^2|\mathbf{b}|^2 + \beta \operatorname{div}\mathbf{\Pi} - \frac{1}{2}m^2\beta^2 \right] \mathrm{d}^3x.$$

Its scalar constraints are $\pi_\beta = 0$ and $\operatorname{div}\mathbf{\Pi} - m^2\beta = 0$.

(a) Eliminate the scalar pair and derive the reduced Hamiltonian and both spatial Hamilton equations with the remaining canonical form. Prove positivity and show that a zero-energy real datum is zero.

(b) At a nonzero Fourier vector k, split the spatial amplitudes into longitudinal and transverse parts. Find the frequencies and a canonical normalisation of each real longitudinal Fourier quadrature. Treat $k = 0$ separately. Why does the energy formula not have a regular mass-zero limit for fixed arbitrary longitudinal momentum?

Individual Fourier modes here analyse the multiplier; they are not claimed to be finite-energy plane-wave fields on $\mathbb{R}^3$.

Hint. Complete the square in β. For the longitudinal part, div contributes a factor $\mathrm{i}|k|$. To normalise an oscillator, rescale its coordinate and momentum by reciprocal factors.

Answer. The reduced energy and equations are

$$H_{\mathrm{red}} = \frac{1}{2}\int \left[|\mathbf{\Pi}|^2 + |\operatorname{curl}\mathbf{b}|^2 + m^2|\mathbf{b}|^2 + m^{-2}|\operatorname{div}\mathbf{\Pi}|^2 \right] \mathrm{d}^3x,$$

$$\mathbf{b}_t = \mathbf{\Pi} - m^{-2}\nabla \operatorname{div}\mathbf{\Pi}, \qquad \mathbf{\Pi}_t = -\operatorname{curl}\operatorname{curl}\mathbf{b} - m^2\mathbf{b}.$$

Every polarisation has frequency $\nu = \sqrt{|k|^2 + m^2}$. For a real longitudinal quadrature, $Q_L = (m/\nu)b_L$ and $P_L = (\nu/m)\Pi_L$ are canonical.